Basic video coding techniques

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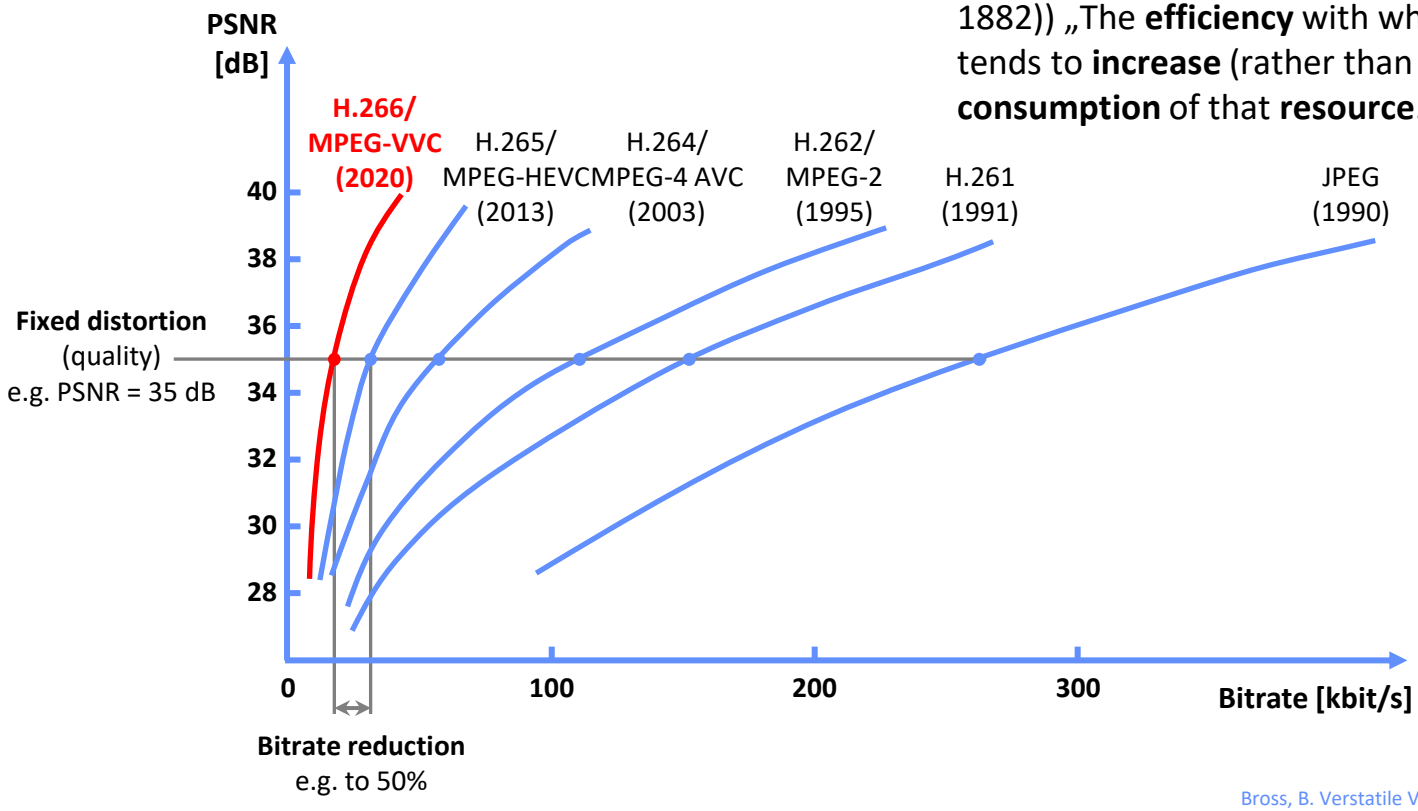
Introduction: Basic motivation

Coding efficiency in standards' generations

Defining requirements for coding efficiency

- **Using RD** (rate/distortion) **curves - bitrate vs. distortion** (quality)
- **Is it still necessary** to **improve coding efficiency?**
  - **Jevons Paradox** (William Stanley Jevons (1835-1882)) „The **efficiency** with which a **resource** is **used** tends to **increase** (rather than decrease) the **rate** of **consumption** of that **resource**."



Bross, B. Verstatile Video Coding (VVC) on the final stretch, 2019.

Introduction

- **Exploiting redundancy** for **video compression**
  - **Four** basic classes of **redundancy** for **video compression**
  - (1) **Temporal** redundancy
    - **Adjacent frames** in sequence **very similar**
    - **Differences** caused by **objects' movement** and/or **camera movement**
      - Utilized in **motion prediction** and **compensation**
  - (2) **Spatial** redundancy
    - High **correlation** among **neighboring pixels**
      - Useful **representation** in (spatial) **frequency domain**
  - (3) **Psychovisual** redundancy (**irrelevancy**)
    - Exploiting **HVS properties** for **lossy compression**
      - Rough **quantization** of **coefficients** for **high spatial frequencies**
  - (4) **Statistical** redundancy
    - Exploiting **relationships among symbols** for **lossless** (entropy) **encoding**

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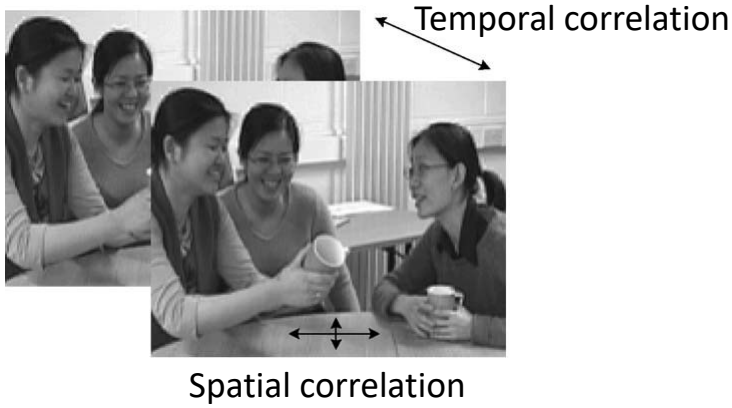
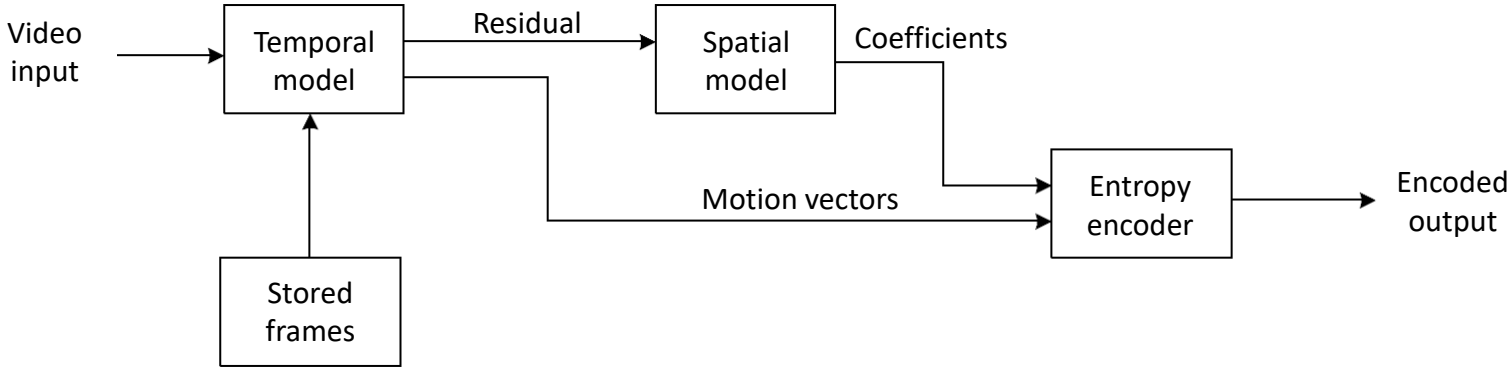
Introduction

- **Exploiting redundancy** for **video compression**
  - **Correlation** (similarity) – **temporal** and **spatial** redundancy



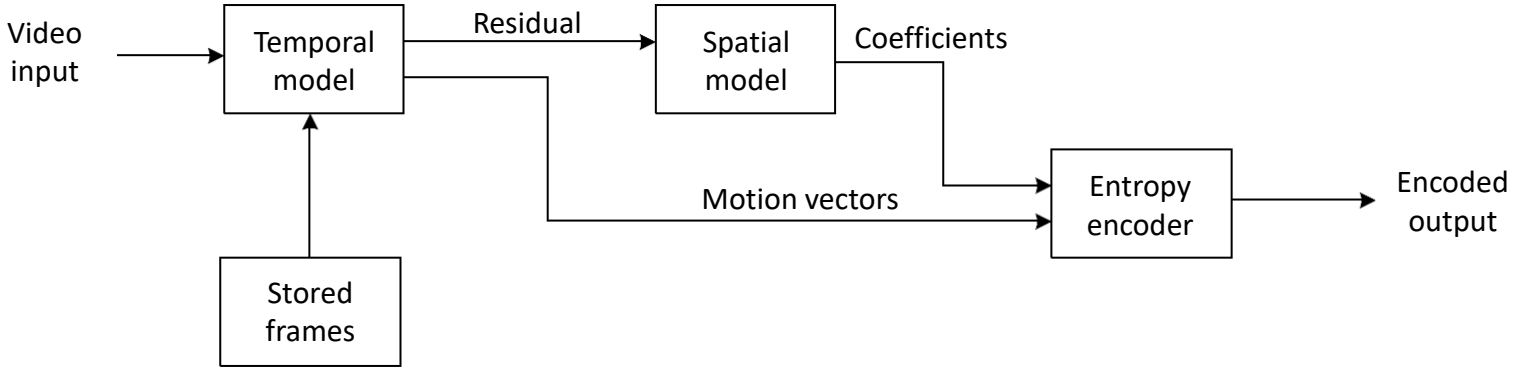
  - Principal block diagram of **video encoder**



Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

Introduction

- **Fundamental blocks** of **general video encoder**
  - **Temporal model** – aiming **temporal redundancy reduction**
    - Exploiting **similarity** between **adjacent video frames**
    - Based on **prediction** of **frame being encoded**
      - Prediction based on **past** and **future** frames with **motion compensation**
  - **Spatial model** – aiming **spatial redundancy reduction**
    - Exploiting **similarity** among **neighboring pixels** in **prediction residual**
    - Based (mostly) on **transform coding** and **quantization**
      - Conversion to **spatial frequency domain** and **coefficients' quantization**



Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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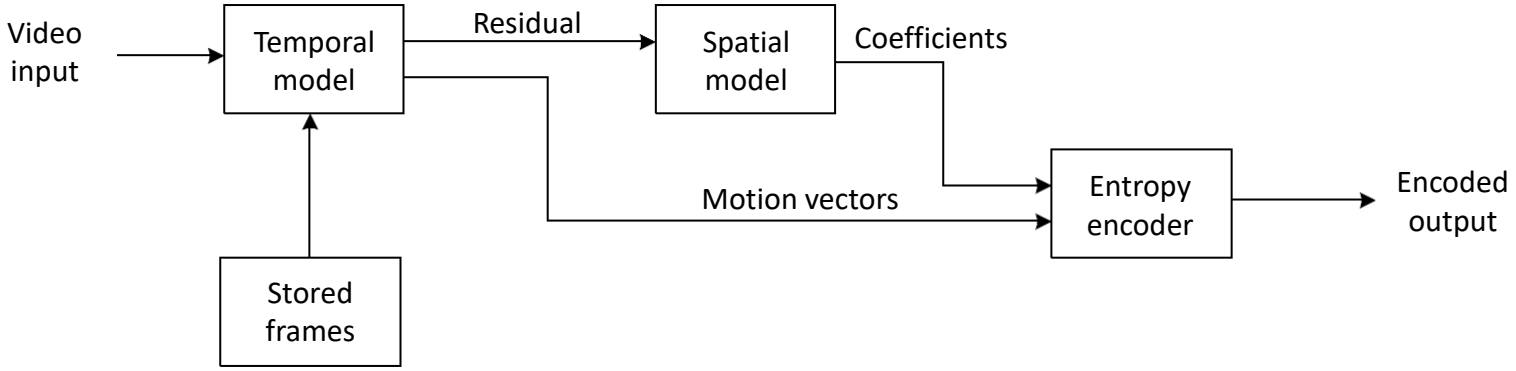
Introduction

- **Fundamental blocks** of **general video encoder**
  - **Entropy encoder** – aiming **statistical redundancy reduction**
    - Exploiting e.g. **different frequencies** of **symbol occurrence**
      - **Temporal model** parameters (**motion vectors**)
      - **Spatial model** coefficients (**quantized coefficients**)
    - Generating **compressed bitstream** for **transmission** or **recording**
      - Compressed bitstream contains mostly **encoded motion vectors**, encoded **prediction residual coefficients** and suitable **header**
- **Inverse procedure applied** in **general video decoder**



Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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Temporal model

- **Temporal redundancy reduction**
  - **Frame prediction** and its **subtraction** from **encoded frame**
    - **Prediction** determined from **one** or **more reference** frames
      - **Reference** frames can be **past** or **future**
    - **Accurate** prediction – **lower energy** (entropy) of **residual frame**
      - **Prediction accuracy** can be **increased** using motion **compensation**
    - Obtaining **prediction error** (**residual frame**)
  - **Reconstruction** in decoder
    - Identical **frame prediction** and **addition** to **residual frame**

- **Simple prediction** from **previous frame**
  - **Efficient** for scenes with **negligible motion**
    - **Residual** frame – high **energy** in case of **prominent scene motion**

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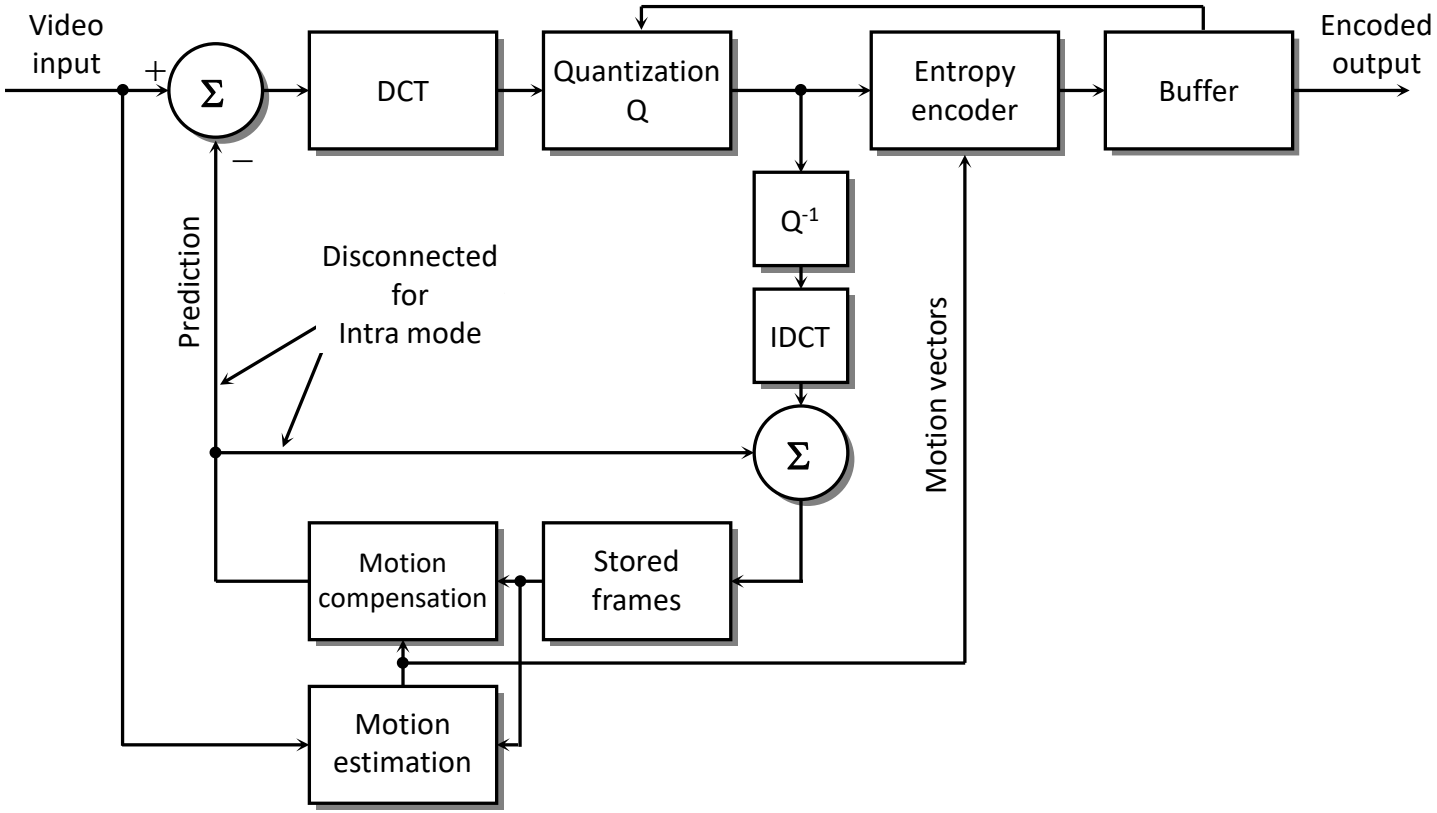
Temporal model

Utilization in hybrid DPCM/DCT encoder



- **Identical general model** for **majority** of **encoders**
  - **Motion compensated prediction** (DPCM), **transform coding**, **quantization** and **entropy encoder**

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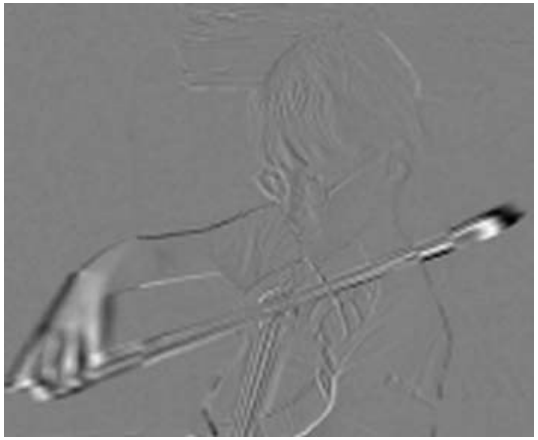
Temporal model

Impact of scene motion

- Simple **prediction** using **previously encoded frame**

Frame S_1

Frame S_2



Difference $S_2 - S_1$

- **Changes between frames**
  - **Objects' motion** in **scene**
  - **Camera motion**
  - **Uncovering** regions
  - Changes in **lighting conditions**
- **Optical flow describes motion of pixels**

Optical flow

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

Temporal model

- **Impact** of **scene motion**
  - **Optical flow** describes **pixels' motion**
    - Allows **accurate prediction** using **pixel shifts**
    - **Rarely used** in video compression
      - Computationally **very complex**
      - Necessary **vector transmission** for **each pixel**

Optical flow

- **Block based motion estimation** and **compensation**
  - Practical use of **motion compensation** for **rectangular blocks**
  - For **each block** $M \times N$ of **currently encoded frame**
    - (1) **Motion estimation** – **search area** in **reference frame** (**past** or **future** frame, previously (de)**coded** and **transmitted**) and **determination of candidate region** $M \times N$
      - Based on **comparing** block $M \times N$ of current **frame** and **all possible blocks** $M \times N$ in **search area** (usually around current block)
      - Criteria for **minimizing residual energy**

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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Temporal model

- (2) **Motion compensation** – **selected candidate** becomes **predictor** for current block $M \times N$
  - **Prediction subtracted** from **current block** obtaining **residual block**
- (3) **Transmission** of **motion vector** and **residual frame**
  - **Entropy encoding** and **transmission** of **residual block** and **motion vector** between **encoded** and **selected** block

❖ **Reconstruction** on **decoder** side

- Reconstruction of **residual block**
- **Prediction** determined based on **motion vector**
- **Adding prediction** to **residual** while obtaining **reconstructed signal**

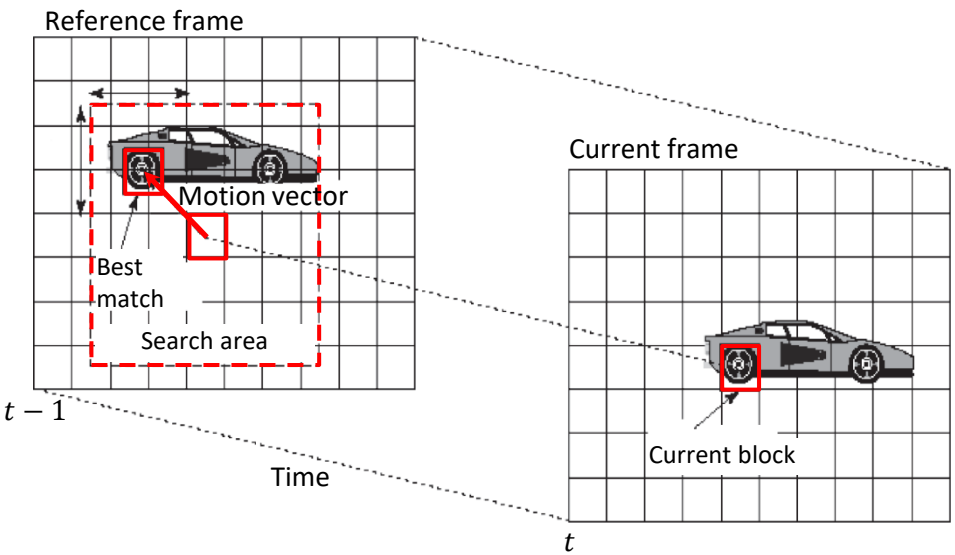
❖ **Advantages** and **disadvantages** of **block based approach**

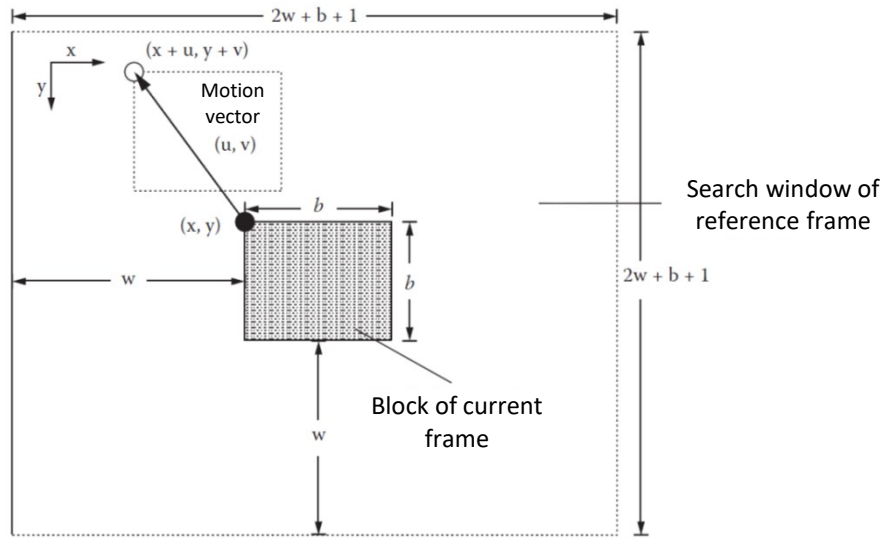
- **Computationally** relatively **simple**
- **Corresponds** to rectangular **frame dimensions** and **DCT** blocks
- Shapes of **real objects** do not match **block shape**
- Real **motion** does not **correspond** to **integer** pixel multiples

Temporal model

Basics of motion estimation

Block based motion estimation





- **Frame** with size $M \times N$ **divided** into **square blocks** $B(i,j)$ with size $(b \times b)$, $i = 1, \dots, M/b$ and $i = 1, \dots, N/b$
- For **each square block** B_m is searched for **region** with **best match**
- **Motion vector** $\forall \vec{r} \in B(i,j)$, $\vec{d}(\vec{r}) = \vec{d}(i,j)$ assigned to all **pixel blocks**, where intensity at position $\vec{r} = (d_x, d_y)^T$ and in time t is $I(\vec{r}, t)$, $\vec{d} = (d_x, d_y)^T$ is shift during time Δt

Raval, M. S., Madhuri A. J., Shilpa P. M., Kalyani R. J., Yogesh H. D.: Image and Video Compression, CRC Press, 2014.

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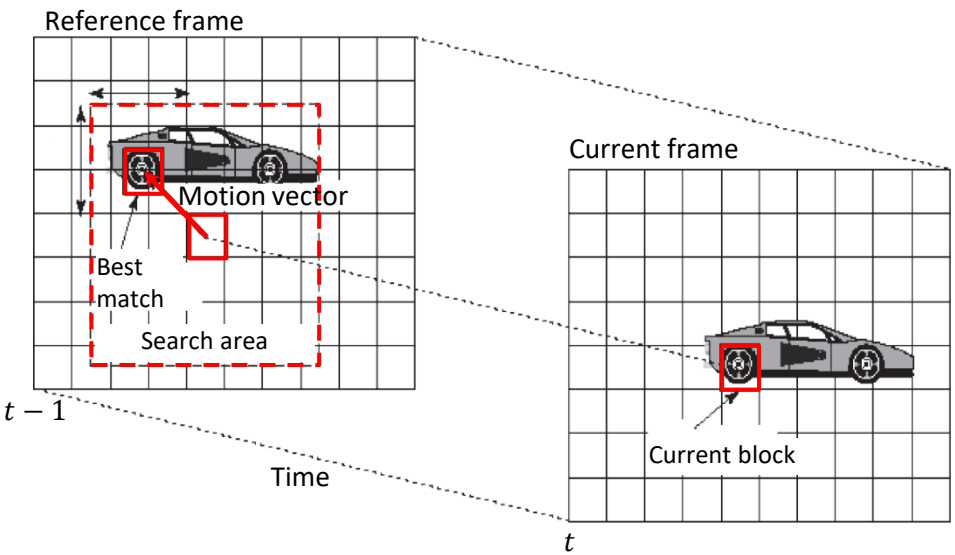
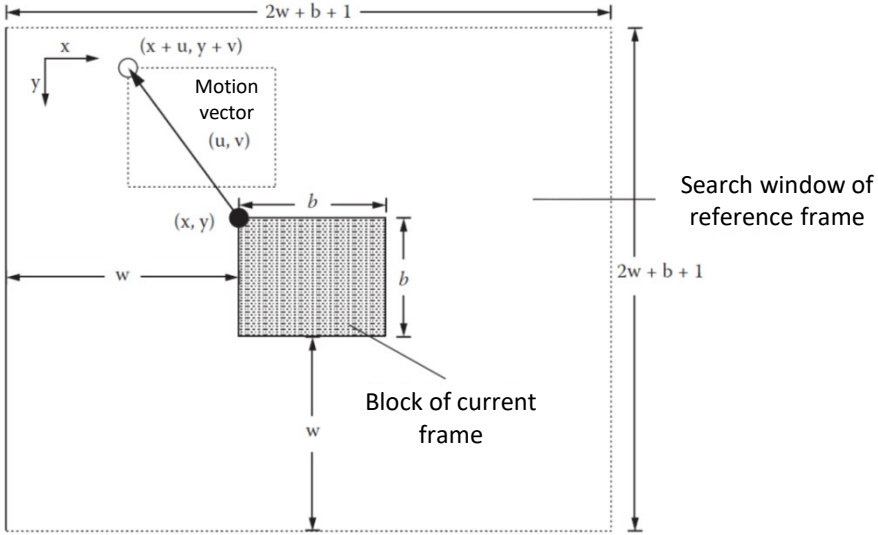
Temporal model

Basics of motion estimation

Block based motion estimation





- If **maximal shift** is $\pm w$ pixels **horizontally** and **vertically**, $(2w+1)^2$ **candidates** in **search window**
- **Global optimum** of selected **metric** can be found with **full** (**exhaustive**) **search**

- Impact on **estimation** and **compensation efficiency** have **distortion metric**, **block size**, size of **search window** and **search method**

Raval, M. S., Madhuri A. J., Shilpa P. M., Kalyani R. J., Yogesh H. D.: Image and Video Compression, CRC Press, 2014.

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Temporal model

Basics of motion estimation

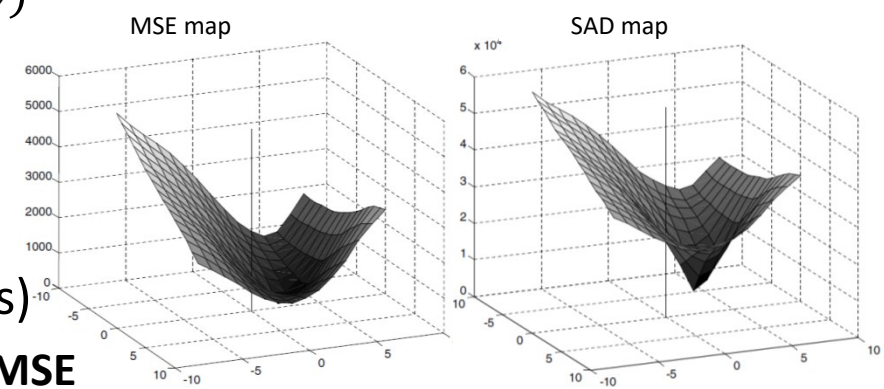
Common metrics for motion vectors' determination

- **Mean square error**
  (MSE – Mean Square Error)

$$MSE(w_x, w_y) = \frac{1}{N^2} \sum_{i=x}^{x+N-1} \sum_{j=y}^{y+N-1} \left(F_t(i,j) - F_{t-1}(i+w_x, j+w_y)\right)^2$$

  – Determination of motion vector (u, v)

$$(u, v) = \min_{-w \le w_x, w_y \le +w} MSE(w_x, w_y)$$



- **Sum of absolute differences**
  (SAD – Sum of Absolute Differences)
  – **Computationally less complex than MSE**

$$SAD(w_x, w_y) = \sum_{i=x}^{x+N-1} \sum_{j=y}^{y+N-1} \left|F_t(i,j) - F_{t-1}(i+w_x, j+w_y)\right|$$

  – Determination of motion vector (u, v)

$$(u, v) = \min_{-w \le w_x, w_y \le +w} SAD(w_x, w_y)$$

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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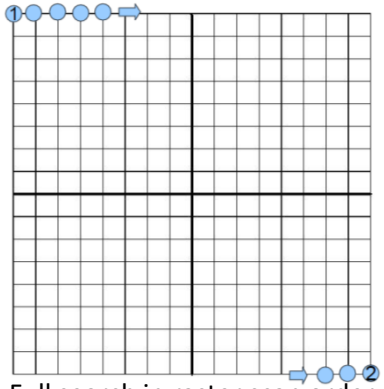
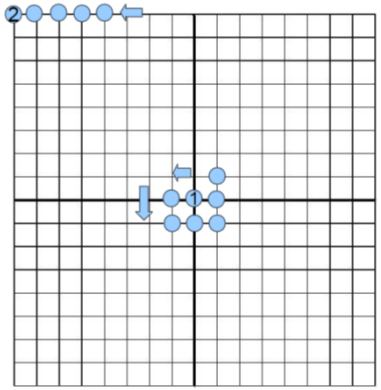
Temporal model

Search algorithms

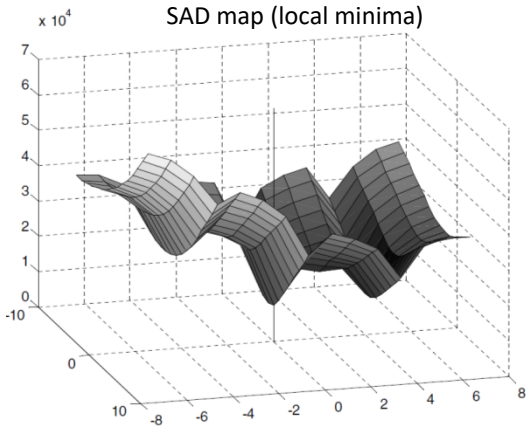
- Procedure for **block search** to **minimize error**
  - Usually **not included** in compression **standard specification**
  - **Global error minimum search** at expense of **computational complexity**
    - **Full** (**exhaustive**) **search** vs. **fast search** (optimized approaches)
- **Full (exhaustive) search (FS)**
  - For each block **search all** $(2w+1)^2$ **positions** in **search window**
  - Computationally **very expensive** – suitable for **HW implementation**
  - Using **raster** or **spiral** search – more common spiral (assigned **shorter vector with minimal distance**)



Full search in raster scan order



Full search in spiral order



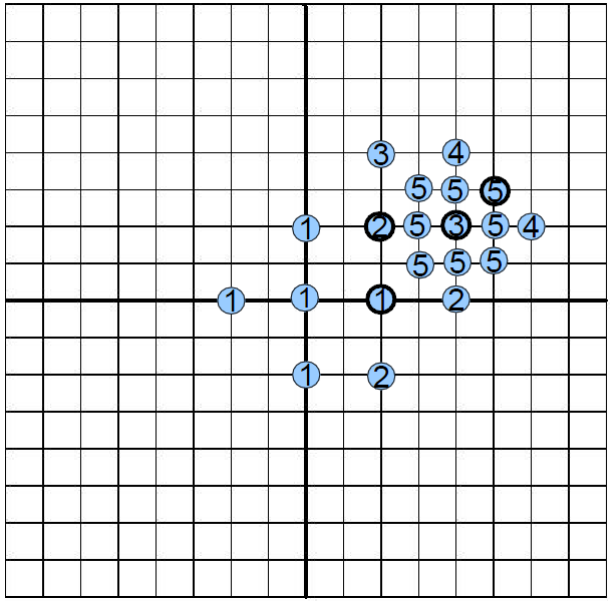
Nešpor, Z.: Techniky pro detekci a predikci pohybu ve video sekvencích, VUT v Brně, 2011.

Temporal model

Search algorithms

Two-dimensional logarithmic search (TDLS)

- **Assumption** – **monotonous error increase** from optimal position
- Initial step $[w/4]$ in 5 positions $[0,0]$, $[0,+s]$, $[0,-s]$, $[-s,0]$, $[+s,0]$
- In case of **smallest error** in $[0,0]$ then **halved step** distance
- Last step **search** in **8 positions** with **step 1**



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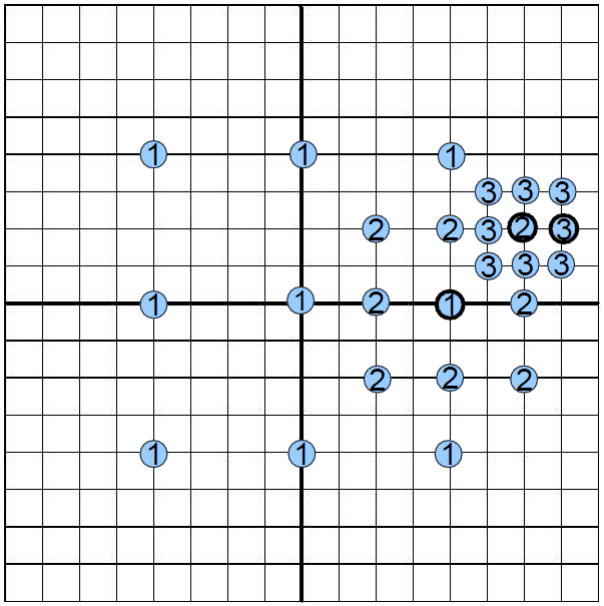
Temporal model

Search algorithms

- **Three step search** (TSS, 3SS)
  - Based on **rough** to **fine** search procedure
  - Search performed for **9 positions** around **origin**
  - Then around **position** with **minimal error** search in **9 positions** with **halved step**
  - Last step **search** in **9 positions** with **step 1**
  - **Limited efficiency** for **slow motion** estimation



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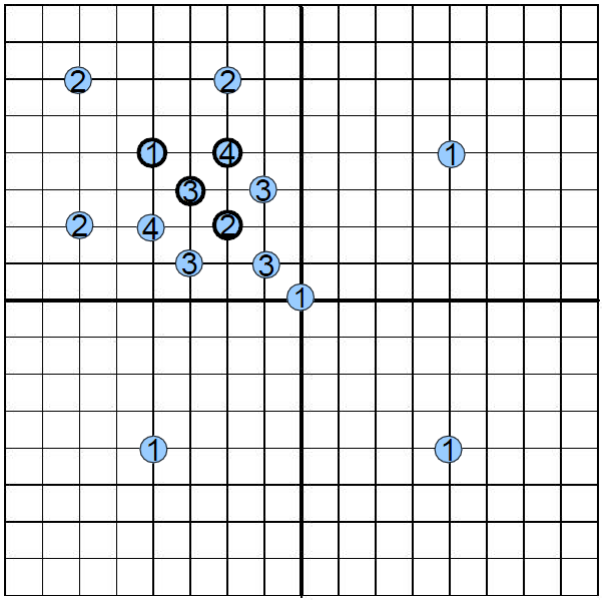
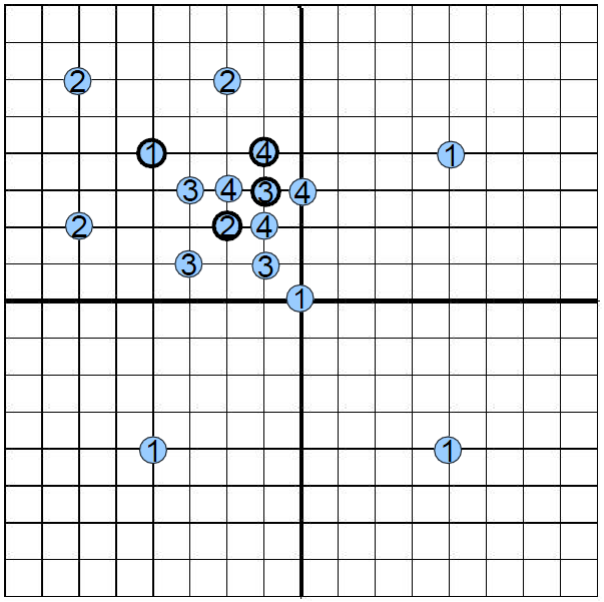
Temporal model

Search algorithms

Cross search algorithm (CSA)

- **Logarithmic search** with **cross pattern**
- Search performed for **5 cross positions**
- Then around **position with minimal error** search in **5 positions** with **halved step**
- Last step **search** in **5 positions** with **step 1** (**two** cross **variants**)
- **Fast** but **lower accuracy**





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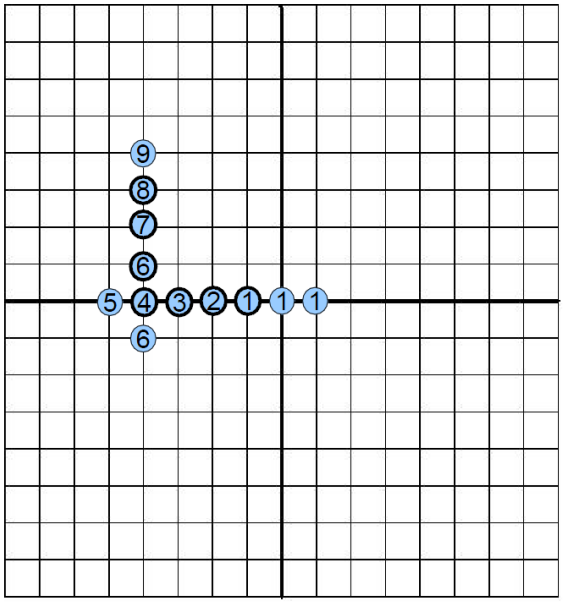
Temporal model

Search algorithms

One at a time search (OTA)

- Simple and **efficient algorithm** with **horizontal** and **vertical** phase
- Search performed with **step 1** in **horizontal** direction
- If optimum in **central position** – then **vertical search**
- Algorithm stopped at **central point** where **lower error** than for neighbors
- **Lower quality/precision** of **motion estimation**



Nešpor, Z.: Techniky pro detekci a predikci pohybu ve video sekvencích, VUT v Brně, 2011.

Temporal model

Search algorithms

Comparison of algorithms' efficiency

- Only **full (exhaustive) search is optimal (global minimum)**
- Fast algorithms **suboptimal** (can identify **local minimum**)
  - Significant **savings** in **required operations**

Algorithm	Average MSE [-]	Average PSNR [dB]	CPU time [s]
FS	12.54	37.86	15.70
3SS	14.84	37.31	1.15
OTA	27.57	36.29	0.37
CSA	18.44	36.41	0.48
TDLS	15.97	37.11	1.43
FS	17.10	38.14	15.70
3SS	18.86	37.88	1.26
OTA	24.89	37.53	0.27
CSA	22.88	37.59	0.44
TDLS	20.05	37.91	1.39
FS	26.36	34.22	15.70
3SS	34.13	33.27	1.11
OTA	42.49	31.59	0.50
CSA	50.98	30.79	0.52
TDLS	32.06	33.44	1.58

Raval, M. S., Madhuri A. J., Shilpa P. M., Kalyani R. J., Yogesh H. D.: Image and Video Compression, CRC Press, 2014.

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Temporal model

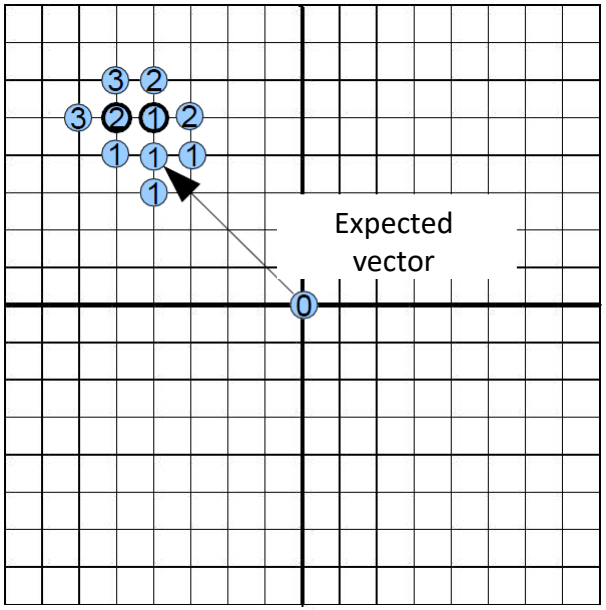
Search algorithms

Hierarchical search

- Search in **downsampled frame versions**
  - Motion vectors calculated for **smallest downsampled frame** (rough estimate)
  - **Vector estimate** used as **initial estimate** at **higher resolution**
  - **Repeated procedure** until **original resolution** reached

Nearest neighbors

- Each **vector** is **estimated** from **neighboring vectors** (already encoded)
  - Exploiting **similarity** in **neighboring vectors**
  - **Cross search** used for **refinement**
  - **Search stopped** if **better match for center point** or when **edge reached**

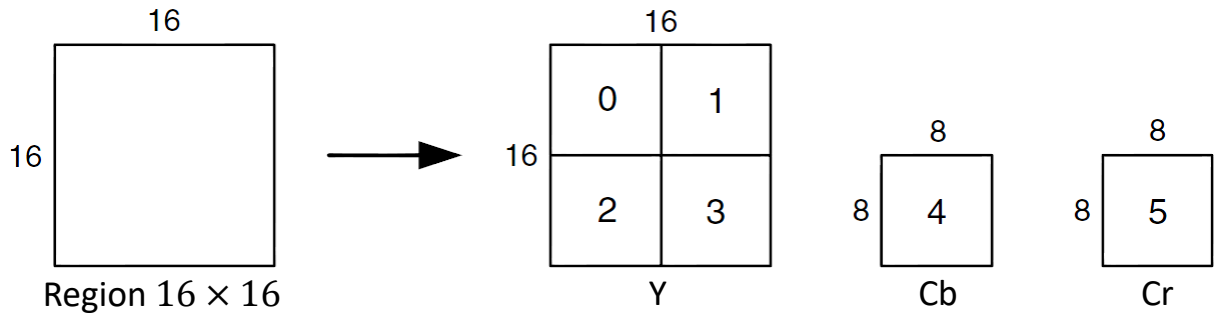


Nešpor, Z.: Techniky pro detekci a predikci pohybu ve video sekvencích, VUT v Brně, 2011.

Temporal model

Motion compensated macroblock prediction

- **Macroblock - basic unit** for **motion compensation**
  - Used in several **standards** MPEG-1, MPEG-2, H.264/AVC,...
  - For **example** in **color video** 4:2:0 **macroblock contains**
    - 16×16 luma and 8×8 chroma samples for Cb and Cr, total **6 blocks** 8×8



- **Motion estimation**
  - Search for **region** with size 16×16 pixels in **reference frame**
    - Block 16×16 – **best matching encoded macroblock** in **search region**
- **Motion compensation**
  - Best **matching region** of reference frame **subtracted**
    - Encoding **residual macroblock** (luma and chroma) and **motion vector**

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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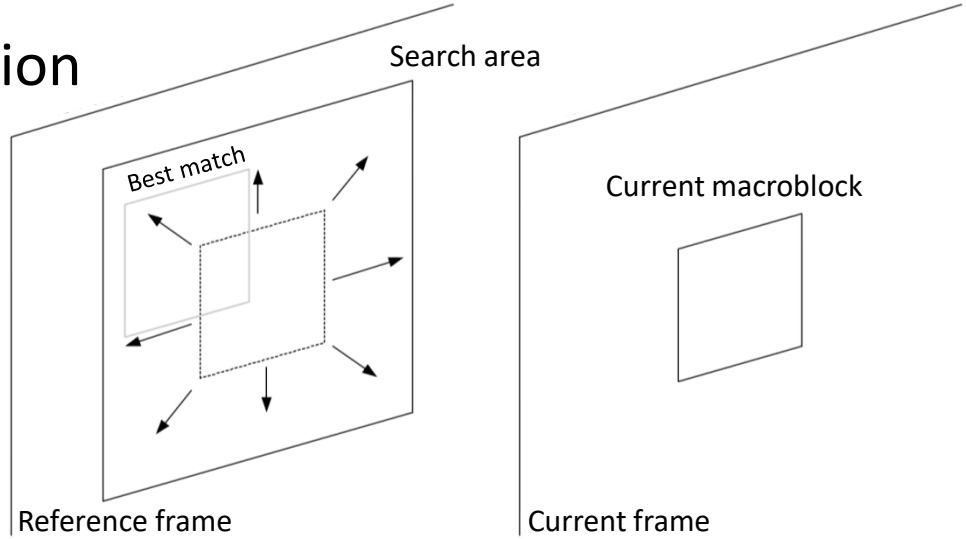
Temporal model

Motion compensated macroblock prediction

- Different **variants** of basic **estimation** and compensation
  - **Reference** can be **past**, **future** frame, or **combination** of **several frames**
  - For **reference** from **future** – **prior** **encoding** (out of time order)



  - Significant **change between frames**
    - Choice for **interframe** (inter mode) and **intraframe** (intra mode) encoding
  - **Objects** usually do not **follow block edges**
    - **Variable block size** can be used for **estimation** and **compensation**
  - **Objects** usually **translate** with **fractional number of pixels** (non-integer)
    - More **accurate prediction** using **reference frame interpolation**

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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Temporal model

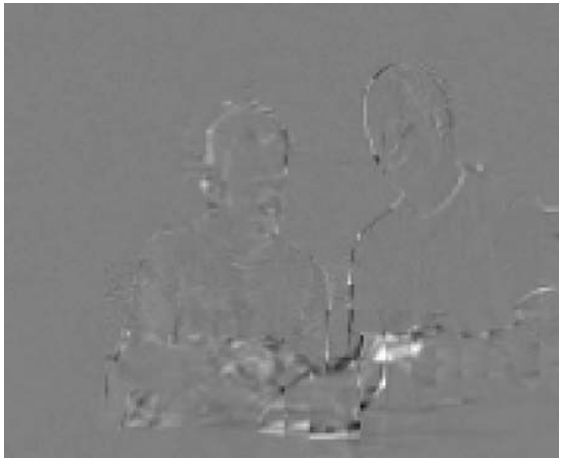
- **Block size impact** on **motion compensation**
  - **Block size** influence on **prediction accuracy**

Frame S_1

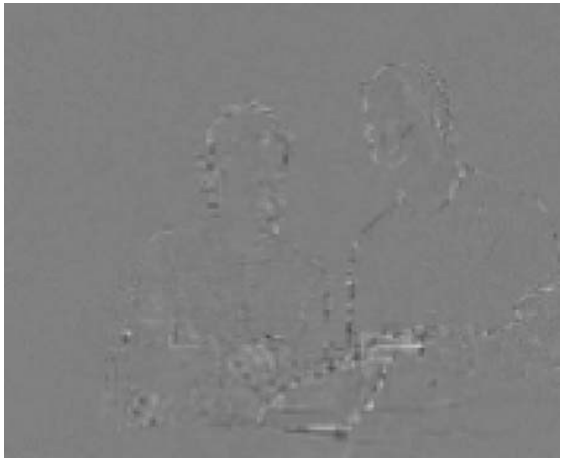
Frame S_2

Residual $S_2 - S_1$

Compensation for blocks 16×16



Compensation for blocks 8×8



Compensation for blocks 4×4

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

Temporal model

Block size impact on motion compensation

- **Block size** influences **prediction accuracy**
  - **Smaller blocks** – **lower energy** (**entropy**) of residual frame
  - **Smaller blocks** – **computational more complex**
    - Need for **more search** operations
  - **Smaller blocks** – **more motion vectors** for transmission
    - Advantage **lower energy** – disadvantage **more bits allocated** to **motion vectors**

- **Block size** can be adapted to **image characteristics**
  - **Large blocks** in **flat/homogeneous regions**
  - **Small blocks** in regions with **details** and **prominent motion**

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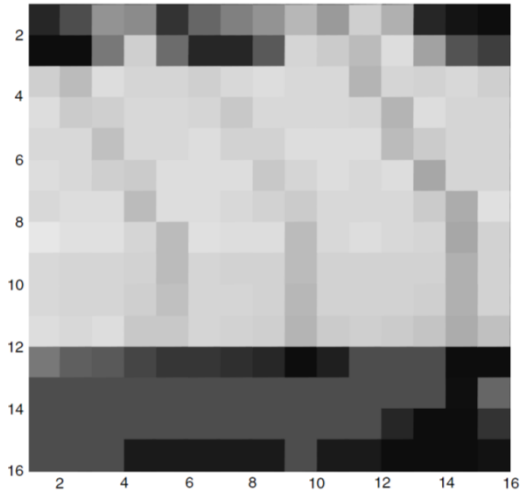
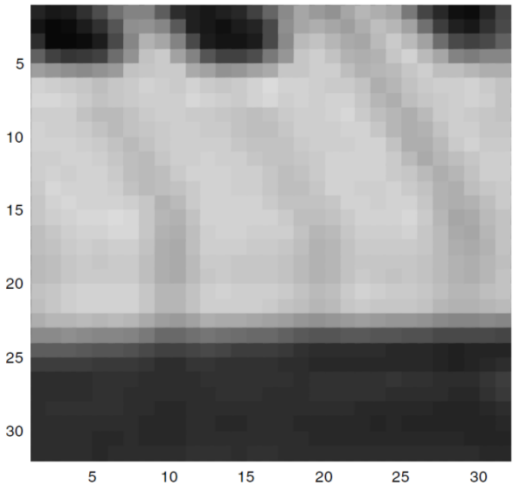
Temporal model

Motion compensation with subpixel accuracy

- Better **motion compensation – interpolated reference**

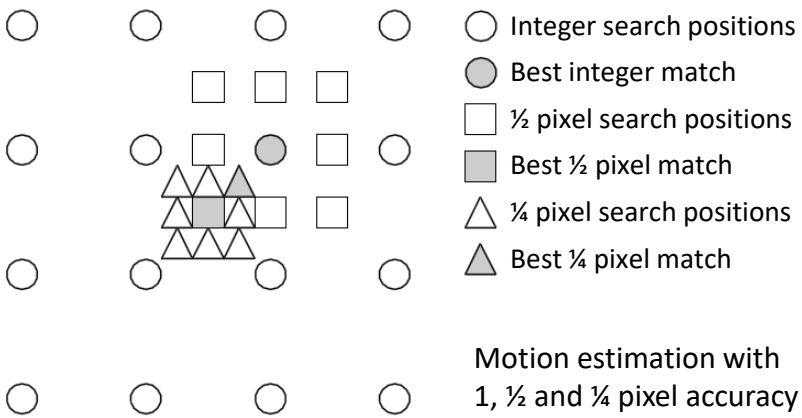


Close-up of reference region



Interpolated to ½ pixel positions

- Search performed with **integer pixel** and **subpixel** shifts
- **Improved accuracy** using **interpolation** to **½** or **¼ sample**
- **Gradual accuracy improvement** for **lower computational complexity**



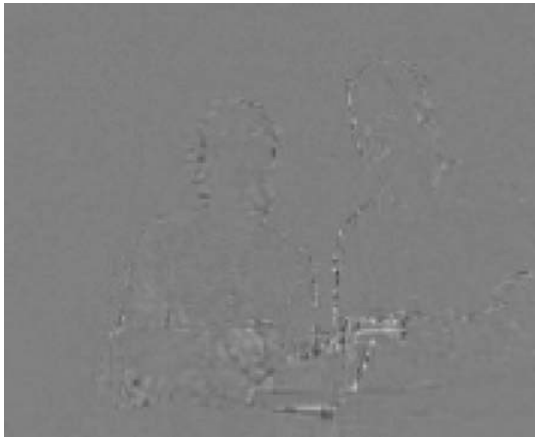
Motion estimation with 1, ½ and ¼ pixel accuracy

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

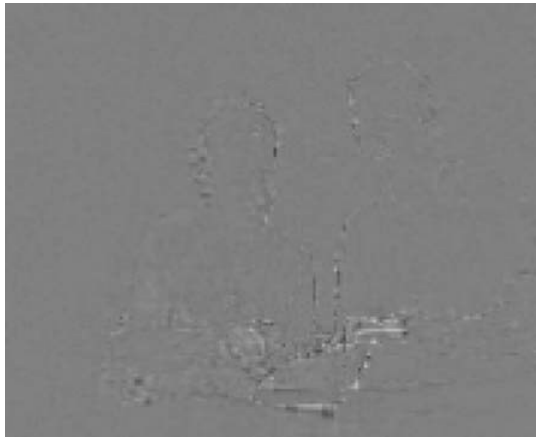
Temporal model

Motion compensation with subpixel accuracy

- Better **motion compensation – interpolated reference**



Residual frame
(blocks 4×4, ½ pixel compensation)



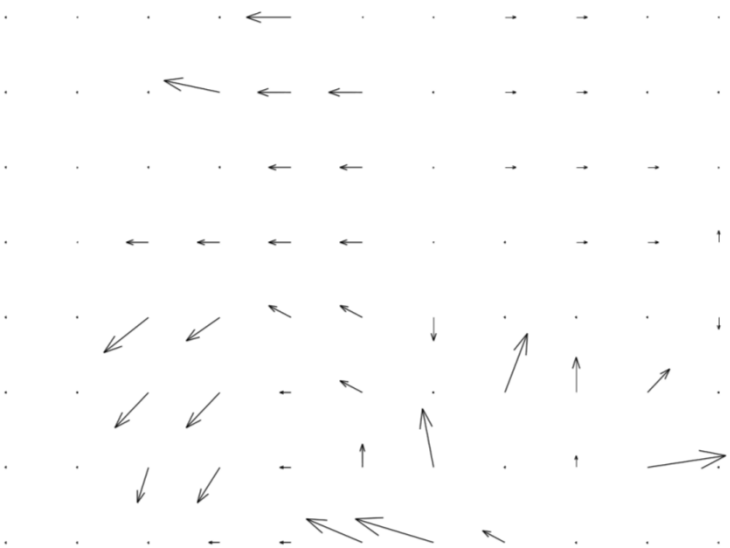
Residual frame
(blocks 4×4, ¼ pixel compensation)

- **Increased computational complexity** and **prediction accuracy**
  - **Prediction residual** decreases **more slowly** with **subpixel accuracy**
  - **Significant improvement** for compensation with **½ pixel accuracy**
  - For **¼ accuracy** and **better – less prominent improvement** in **encoding efficiency**

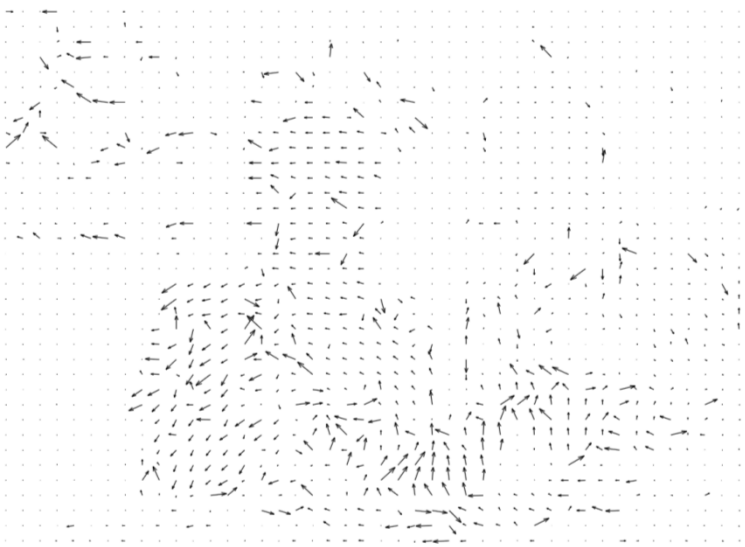
Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

Temporal model

- **Effect** of **smaller blocks** and **subpixel accuracy**
  - **Main advantages**
    - Better **compensation** with **lower energy** of **residual frame**
  - **Main disadvantages**
    - **Computational demands** and **amount** of **auxiliary data**
      - For blocks 4×4 - $\mathbf{16} \times$ **more vectors** than for blocks 16×16



Example of motion vector map
(16×16 blocks, integer vectors)



Example of motion vector map
(4×4 blocks, ¼ pixel vectors)

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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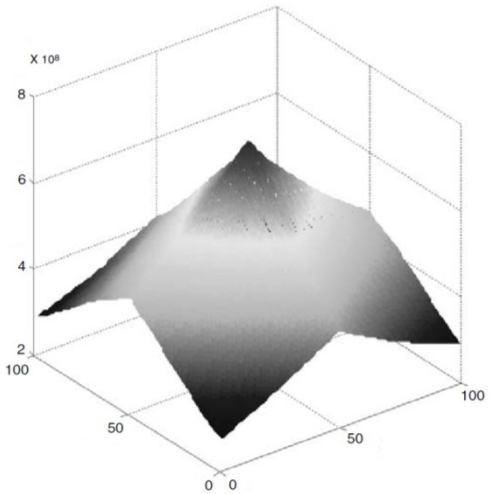
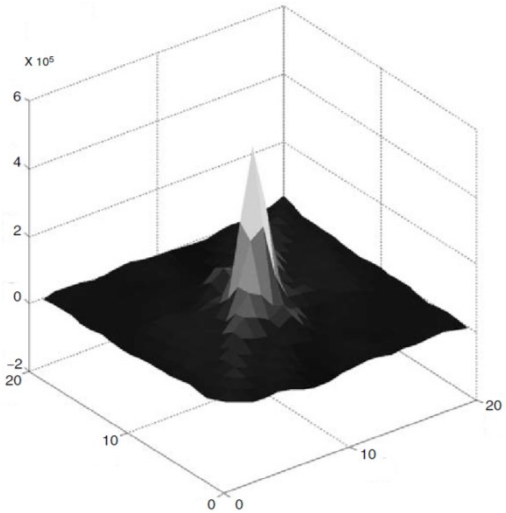
Spatial model

- **Exploiting correlation among neighboring pixels**
  - Relationship described with **2D autocorrelation function**
    - Describes **similarity** of **frame** and its **spatially shifted replica**



Typical image 2D autocorrelation function



Typical residual 2D autocorrelation function

    - **Motion compensated residual frame** has **lower correlation**
  - **Spatial model** further **decorrelates frame** (residual frame)
    - **Residual** can be **efficiently encoded** using **entropy encoder**

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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Spatial model

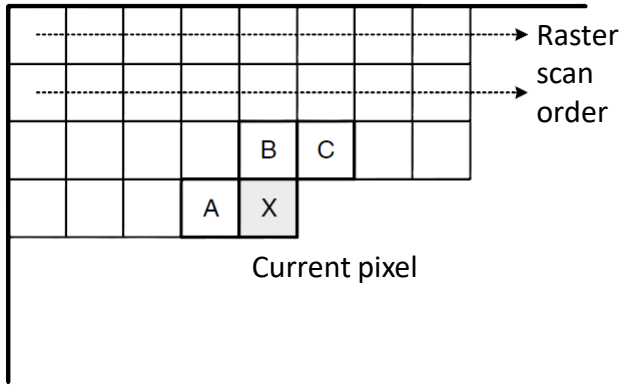
- **Three main components of spatial model**
  - (1) **Transform** – data **decorrelation**, **compact** representation
  - (2) **Quantization** – **accuracy reduction** of transform **coefficients**
  - (3) **Reordering** – ordering **according** to **significance** into groups
- **Predictive coding**
  - **Predictive coding** for **interframe prediction** (inter)
    - **Interframe prediction**
    - **Motion estimation** and **motion compensated prediction** (motion compensation)
  - **Predictive coding** for **intraframe prediction** (intra)
    - **Spatial prediction** – differential pulse code modulation (**DPCM**)

Spatial model

Predictive coding

Predictive coding for intraframe prediction

- **Spatial prediction** – difference pulse code modulation (DPCM)
- Prediction of **current pixel** (X) using **neighboring** pixels available in **encoder** and **decoder** (A, B, C)
- Coder determines **prediction** based on **combination** of encoded pixels (A, B, C)
- **Prediction** is **subtracted** from **current encoded pixel** (X)
- Results in **residual pixel** (prediction error)
- Decoder **calculates prediction** and **adds** to **received residual**
- **Example**
  - Prediction $P(X) = (2A + B + C)/4$, residual $R(X) = X - P(X)$
  - Decoder receives $R(X)$, determines prediction $P(X) = (2A + B + C)/4$
  - Pixel reconstruction $X = R(X) + P(X)$



Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

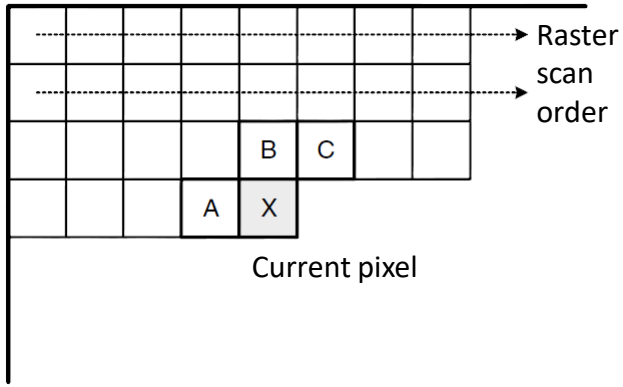
Spatial model

Predictive coding

Modification for lossy predictive coding

- For **quantized residual decoded pixels** do **not necessarily match original**
  - A', B' and C' **do not match original** A, B and C
  - Results in **error accumulation** (drift) **between encoder** and **decoder**
- Encoder **reconstructs** quantized residual $R'(X)$ and **every pixel**
- **Prediction** on **encoder side** $P(X)$ from **decoded pixels** A', B' and C'

$$P(X) = (2A' + B' + C')/4$$



- Encoder and decoder **use identical prediction** and **prevents drifts**

Compression efficiency of predictive coding

- Depends on **prediction accuracy** $P(X)$
  - **More accurate approximation** X → **small residual energy**
- Higher **prediction accuracy** achieved by **adaptive predictors**

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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Spatial model

Transform coding

- **Conversion** from **pixel** to **transform** domain
  - Transformation of **frame** or **residual** after **predictive coding**
- **Requirements** on **transform**
  - (**1**) **Coefficients** of transform domain **decorrelated** (minimal inter-dependence) and **compact representation** (small number of coefficients)
  - (**2**) **Transform** should be **reversible**
  - (**3**) **Transform** should be **computationally inexpensive** (low memory requirements, limited-precision arithmetic, low number of operations,...)
- Popular **transform groups**
  - **Block-based transforms** (DCT, KLT,...) – processing in **blocks** $N \times N$
    - Low **memory requirements**, **residual transformation** after motion compensated prediction, block artifacts,...
  - **Image-based transforms** (DWT,...) – processing of **entire image** or **tiles**
    - Higher **memory requirements**, **better efficiency** than DCT..., **less suitable for residual transform** after motion compensated prediction,...

Spatial model

Discrete cosine transform (DCT)

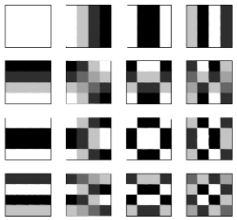
- Most common **transform** for **video compression**

$$\mathbf{Y} = \mathbf{A}\,\mathbf{X}\,\mathbf{A}^{\mathrm{T}}$$
$$\mathbf{X} = \mathbf{A}^{\mathrm{T}}\mathbf{Y}\,\mathbf{A}$$

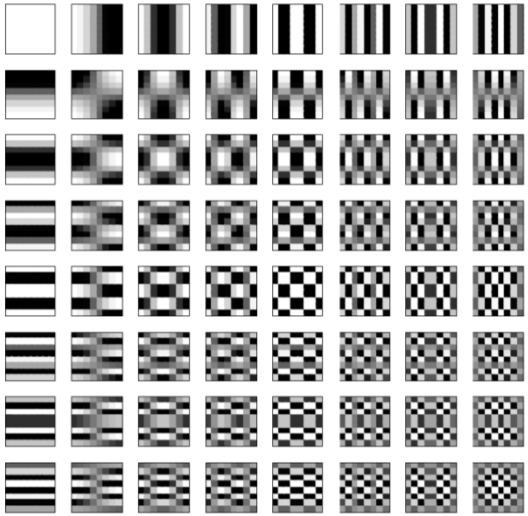
where $\mathbf{X}$ is pixel matrix, $\mathbf{Y}$ coefficients and $\mathbf{A}$ is transform matrix $N \times N$

$$A_{ij} = C_i \cos\frac{(2j+1)i\pi}{2N}, \qquad C_{i=0} = \sqrt{\frac{1}{N}}, \qquad C_{i>0} = \sqrt{\frac{2}{N}}$$

- **Block decomposition** into **basis patterns**
  - Block reconstructed as **linear combination** of **basis patters**



Basis patterns for 4×4 DCT



Basis patterns for 8×8 DCT

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

Spatial model

Quantization

- **Quantizer maps** signal X to quantized signal Y
  - Signal Y with **smaller number of values** than signal X
  - **Scalar quantizer** more **common** in **video compression**

- **Quantization** as **lossy procedure** (irreversible)
  - Possible **implementation** using **rounding**

$$FQ = \text{round}\frac{X}{QP}, \qquad Y = FQ \cdot QP$$

    where QP is **quantization step**
  - Encoder contains **forward quantizer** (FQ) and decoder **inverse quantizer** (IQ)
  - Important selection of **quantization step** QP (controls **amount** of **loss**)
  - Used to **lower representation accuracy** of **coefficients** (e.g. DCT)
  - Number of **small coefficients** turn to **zero**

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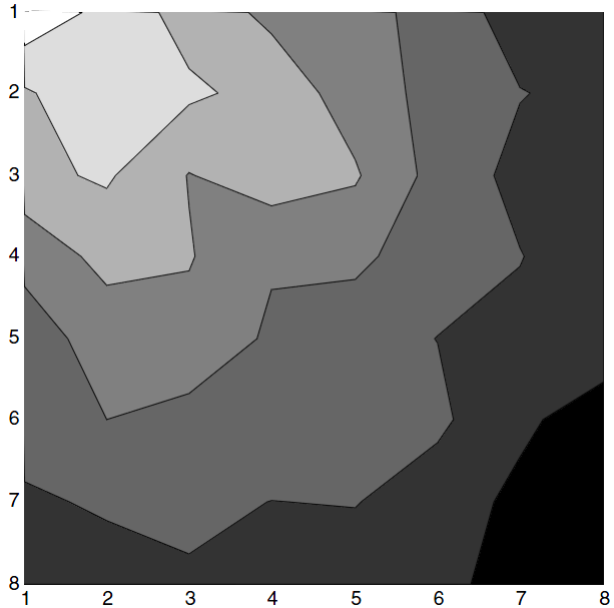
Spatial model

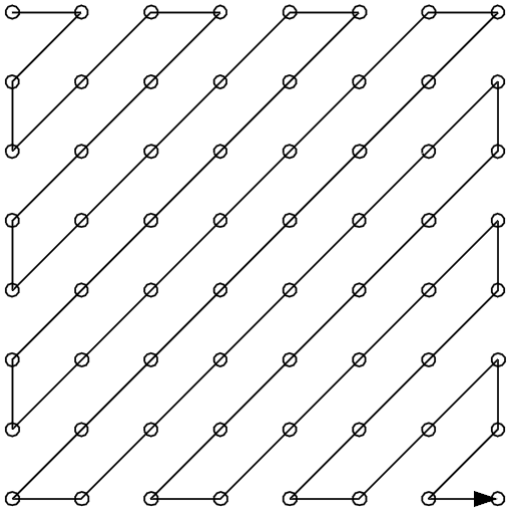
Quantization and DCT

Quantized transformation coefficients

- **Sparse matrix** of transform **quantized coefficients**
- Few **nonzero coefficients** and **large number** of **zeros**
- **Reordering** and **efficient representation** of **zero coefficients**
- **Significant DCT coefficients around DC** $(0,0)$ **– zig-zag readout**



Typical distribution of 8×8 DCT coefficients



Zig-zag readout of 8×8 DCT coefficients

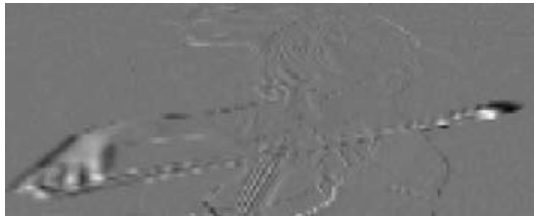
Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

Spatial model

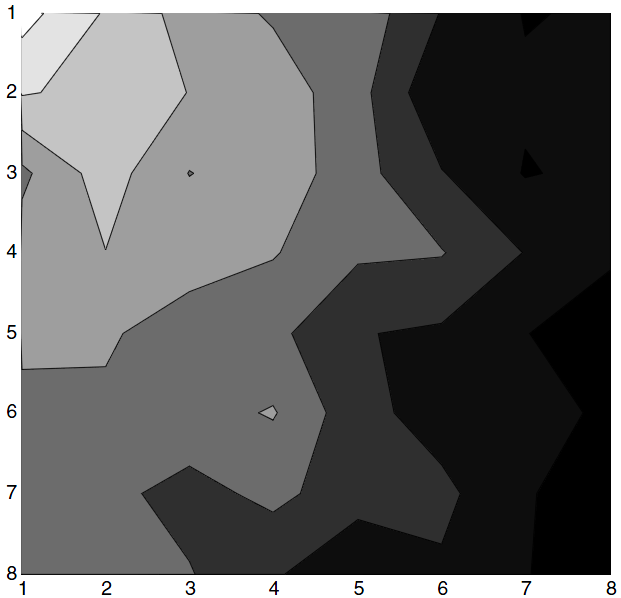
Quantization and DCT

Quantized transformation coefficients

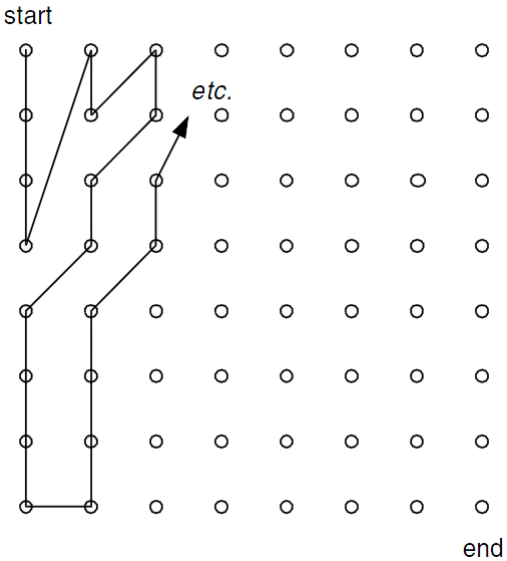
- **Special cases** - encoding **fields** in **interlaced sampling**



Residual for field encoding



Typical distribution of 8×8 DCT coefficients



Zig-zag readout of 8×8 DCT coefficients

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

Spatial model

Quantization and DCT

Run-level encoding

- **RLE** - Run-Level Encoding, Run-Length Encoding
- **Reordered coefficients** (zig-zag) contain **strings** of **zero coefficients**
- **Example**
  - **Input array**: 16,0,0,−3,5,6,0,0,0,0,−7
  - **Output values**: (0,16),(2,−3),(0,5),(0,6),(4,−7)
- Last **nonzero coefficient** followed by **EOB symbol**

Using **3D RLE – last value** in **three quantities** assigned **EOB**

- **Example**
  - **Input array**: 16,0,0,−3,5,6,0,0,0,0,−7
  - **Output values**: (0, 16, 0), (2,−3, 0), (0, 5, 0), (0, 6, 0), (4,−7, 1)
- Each **pair/triple** encoded using **separate symbol**

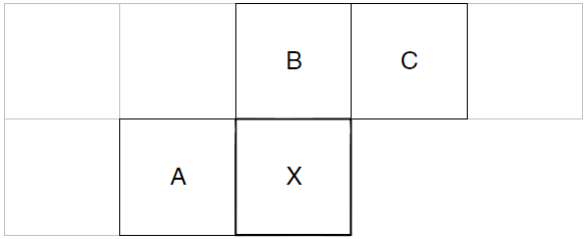
Entropy coder

- **Encoding symbols** into **bitstream**
  - **Symbols** represent **videosequence elements** as follows
    - **Quantized transform coefficients**
      - **Encoded using RLE**
    - **Motion vectors**
      - **Displacements** x and y for **each motion compensated block**
      - **Integer pixel** or **subpixel** accuracy
    - Markers
      - Codes indicating synchronization points in sequence
    - Headers
      - Macroblock headers, picture headers, sequence headers,...
    - Supplementary information
      - Side information not essential for correct decoding
  - Common **entropy coding techniques**
    - **Predictive pre-coding**, **modified Huffman** and **arithmetic coder**, ...

Entropy coder

Pre-coding using predictive coder

- Selected **symbols** are **highly correlated** in regions
  - For instance **average** value (DC - coefficient) of **neighboring** blocks
  - **Displacements** of **motion vector** in **neighboring** blocks
  - Better **compression efficiency** – **prediction** and **difference** transmission
- For **small blocks** (e.g. 4×4) and **large objects**
  - Large **similarity** of neighboring **motion vectors**
  - Simple **prediction** of **motion vectors**
    - Exploiting **information** from **adjacent blocks** (A, or combination A, B, C)
    - Transmission of **difference motion vectors** (MVD – Motion Vector Difference)



- Signaling **quantization parameter QP**
  - **QP variations** to keep required bitrate
  - **QP signalization** to **decoder** – using **difference**, e.g. ± 1, ± 2

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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Entropy coder

Variable-length coding

- VLC – Variable-length Coding
- Maps **every symbol** to **codeword** with **variable length**
  - **Frequent symbols** have **shorter codeword** and opposite

- **Huffman coding**
  - **Variable length code** based on **probability** of **symbol occurrence**
  - For **given probability distribution** is **Huffman code compact**
    - **Adaptation** – **adaptive coding tables** based on distribution
    - Disadvantage of **integer number** of **bits** in codewords
  - **Precalculated Huffman coder tables**
    - Intended for **typically encoded sequences** (suboptimal)
    - Available in **encoder** and **decoder**
  - **Typical tables**
    - For **transform coefficients**, **motion vectors**, ...

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Entropy coder

Variable-length coding

Huffman coding

- **Example table** (MPEG-4 Visual) for **transform coefficients** and **differential motion vectors**
  - Selected **3D-RLE coefficients** (TCOEF) encoded using VLC, otherwise 13 bit code (Escape)
  - Table for **MVD**

VLC for Motion Vector Difference (MVD) in MPEG4

MVD	Code
0	1
+0.5	010
−0.5	011
+1	0010
−1	0011
+1.5	00010
−1.5	00011
+2	0000110
−2	0000111
+2.5	00001010
−2.5	00001011
+3	00001000
−3	00001001
+3.5	00000110
−3.5	00000111
...	...

VLC for Visual Transform Coefficients (TCOEF) in MPEG4

Last	Run	Level	Code
0	0	1	10s
0	1	1	110s
0	2	1	1110s
0	0	2	1111s
1	0	1	0111s
0	3	1	01101s
0	4	1	01100s
0	5	1	01011s
0	0	3	010101s
0	1	2	010100s
0	6	1	010011s
0	7	1	010010s
0	8	1	010001s
0	9	1	010000s
1	1	1	001111s
1	2	1	001110s
1	3	1	001101s
1	4	1	001100s
0	0	4	0010111s
0	10	1	0010110s
0	11	1	0010101s
0	12	1	0010100s
1	5	1	0010011s
1	6	1	0010010s
1	7	1	0010001s
1	8	1	0010000s
ESCAPE			0000011s
...	...	...	...

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Entropy coder

Variable-length coding

- **Huffman code** is **not transmission error resilient**
  - **Transmission error** can cause decoder **synchronization loss**
  - There are **other derived error resilient codes**
- **Encoder** and **decoder** contains **predefined coding table**

- **Other variable-length codes**
  - **Reversible VLC** (RVLC)
    - Decodable **forward** and **backward** direction – better **decoding efficiency** with errors
  - **Exponential Golomb** code (Exp-Golomb)
    - Can be **generated automatically** on the fly

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Entropy coder

Arithmetic coding

Huffman code

- **Codeword** has **integer number** of **bits** for each **symbol**
- **Suboptimal** – **theoretically** required **number** of **bits usually fractional**
- **Low efficiency** especially for **symbols with probability higher than 0.5**
  - These symbols are **represented using one bit codeword**

Arithmetic coding

- **Sequence** of **data symbols** encoded **together**
- **Efficiency** close to **theoretical maximum**
  - No limit to **integer number** of **bits** per **symbol**

Context-based arithmetic coder (CAE – context based)

- **Efficiency** of coding depends on **probability model accuracy**
  - **Probability prediction** for **spatial** and **temporal** characteristics

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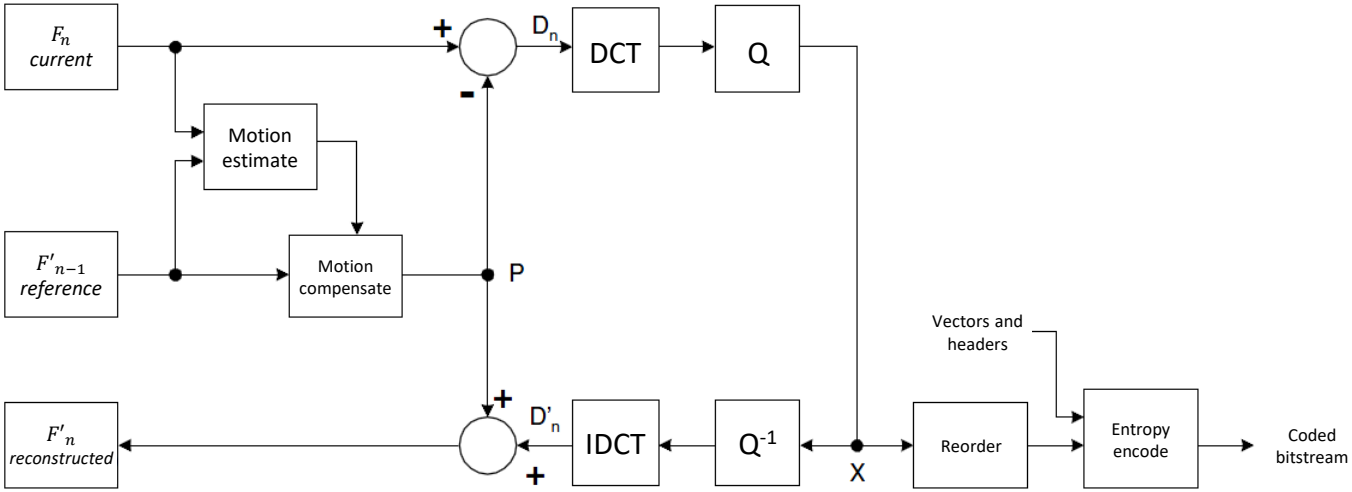
Hybrid DPCM/DCT coder

- **Majority** of **coders based** on **hybrid DPCM/DCT model**
  - **Hybrid DPCM/DCT coder model contains**
    - **Motion estimation** and **motion compensated prediction**
    - **Transform coding**
    - **Entropy encoder**
  - **Almost identical structure for many standards**
    - H.261, H.263, MPEG-1, MPEG-2, MPEG-4 Visual, H.264/AVC, H.265/HEVC,...
      - **Identical model** – numerous differences in respective blocks
  - **Encoder** and **decoder**
    - Can be **described** with **identical general block diagram**
      - n-**th input sequence video frame** (F_n) is encoded into bitstream in encoder
      - Decoder decodes bitstream and **reconstructs frame** F'_n
      - **Lossy compression** - F_n and F'_n are **not identical**

Hybrid DPCM/DCT coder

DPCM/DCT video encoder



- Basic **general block scheme**
  - **Encoder** has two paths – **encoding** (→) and **reconstruction** (←)
  - **General encoding procedure** (→)
    - (**1**) Input **video frame** F_n is processed in **macroblocks** (16×16 luma and corresponding chroma)
    - (**2**) Frame F_n **compared** with **reference** (e.g. previously encoded frame F'_{n-1}) and **region** 16×16 **found** in F'_{n-1} (F'_{n-1} can be interpolated for subpixel accuracy) with **best match** with current macroblock in F_n
    - Displacement signalized with **motion vector** MV – Motion Vector

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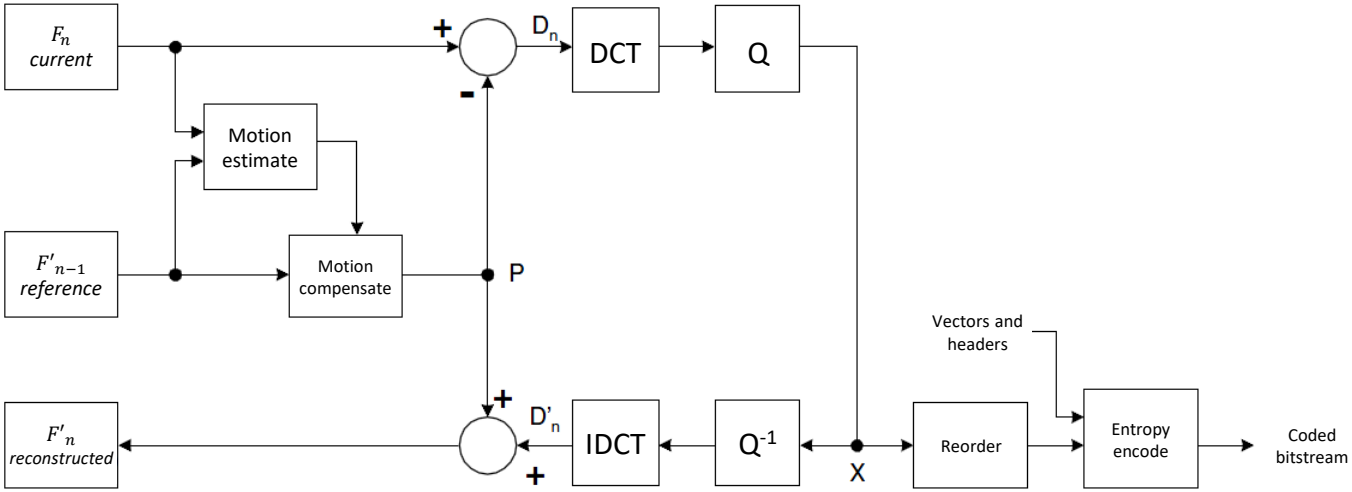
Hybrid DPCM/DCT coder

DPCM/DCT video encoder



- **General encoding procedure** (→)
  - (**3**) Using MV **motion compensated prediction** P is performed – estimator is **selected region** 16×16 in reference F'_{n-1}
  - (**4**) **Prediction P subtracted** from current macroblock → **residual** (differ.) D_n
  - (**5**) **Transformation** D_n **using DCT** – typically D_n divided in **blocks** 8×8 or 4×4 – **transformed independently**
  - (**6**) Each **submatrix of coefficients quantized** (X)
  - (**7**) Quantized **DCT coefficients reordered** and **RLE applied**
  - (**8**) **Entropy encoding** of coefficients, motion vectors, headers,...

Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

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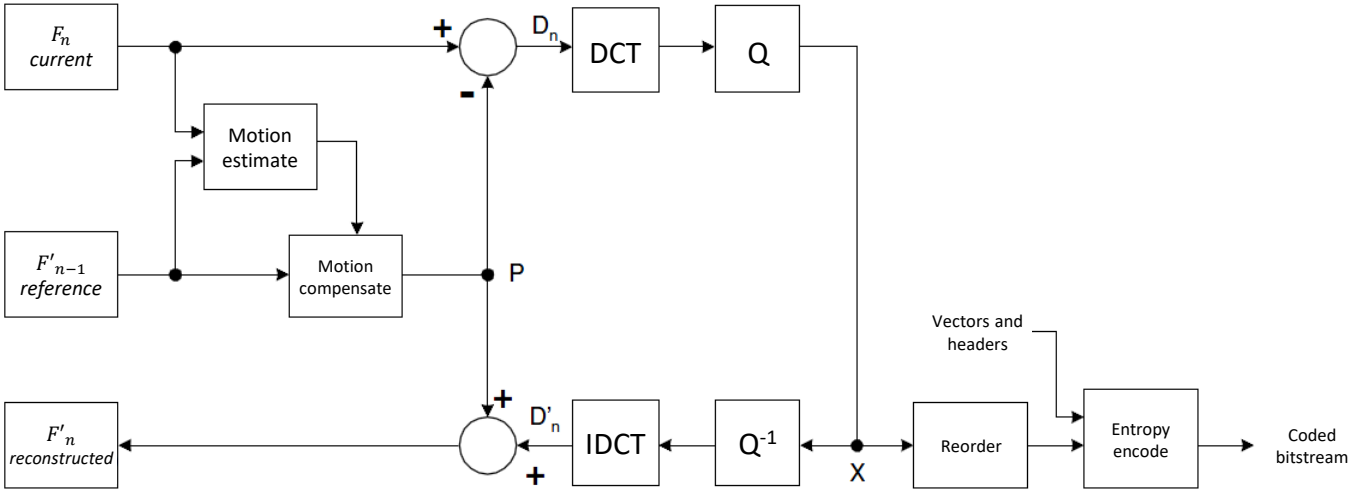

- **General reconstruction procedure** (←)
  - (**1**) For each **quantized macroblock** X **inverse quantization** and **transform** applied. **Decoded residual** D'_n obtained – not identical with D_n (**distortion** due to **lossy encoding** – quantization)
  - (**2**) **Motion compensated prediction** P added to D'_n and determined **reconstructed macroblock** – obtaining **reconstructed frame** F'_n
  - (**3**) **Reconstructed frame** F'_n can be used as **reference** next encoded frame F_{n+1}

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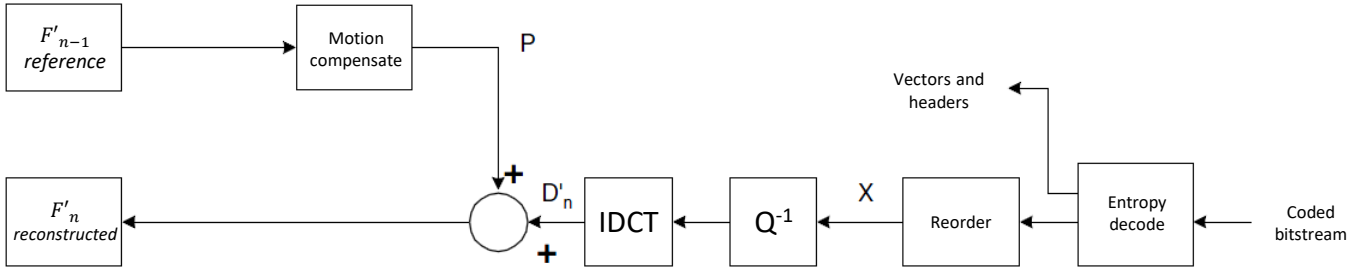
Hybrid DPCM/DCT coder

DPCM/DCT video decoder



- **General reconstruction procedure** (←)
  - (**1**) From **encoded bitstream** entropic decoder extracts **quantized coefficients**, **motion vectors** and headers for **each macroblock**
  - (**2**) **RLE decoding** (reorder) performed and **macroblock of quantized coefficients** X is obtained
  - (**3**) For each **quantized macroblock** X **inverse quantization** and **transform** applied – **decoded residual** (difference) D'_n obtained
  - (**4**) **Motion vector** used to determine **region** 16×16 in **reference frame** F'_{n-1} (**identical** to **encoder**) – **compensated prediction** P
  - (**5**) **Prediction** P added to **decoded residual** D'_n, **reconstructed macroblock** obtained, and **decoded frame** F'_n calculated
  - (**6**) **Decoded frame** F'_n can be **displayed** and saved as **reference** for next decoded frame F_{n+1}

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Hybrid DPCM/DCT coder

Example

Input frame F_n

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Hybrid DPCM/DCT coder

Example

Reconstructed reference frame F'_{n-1}

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Hybrid DPCM/DCT coder

Example

Residual frame $F_n - F'_{n-1}$ (no motion compensation)

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Hybrid DPCM/DCT coder

Example

Motion vectors for macroblocks 16×16 superimposed on frame

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Hybrid DPCM/DCT coder

Example

Motion compensated reference frame F'_{n-1}

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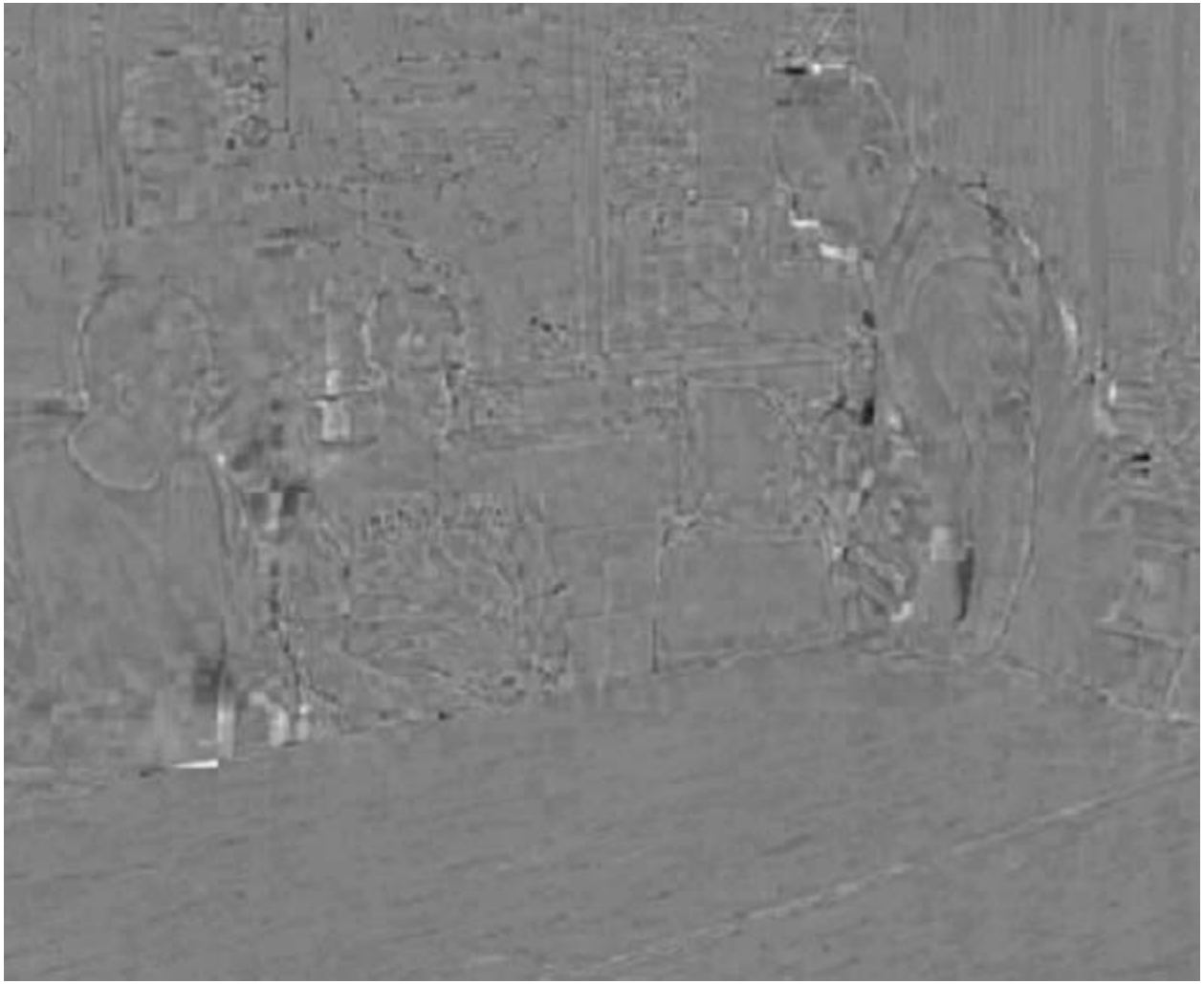
Hybrid DPCM/DCT coder

Example



Motion compensated residual frame $F_n - F'_{n-1}$

No motion compensation

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Hybrid DPCM/DCT coder

Example

Decoded frame F'_n

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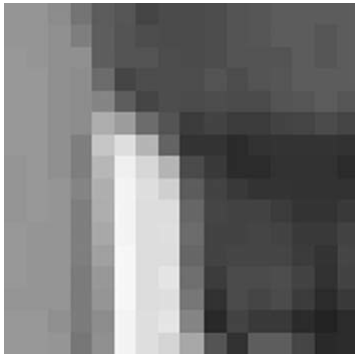
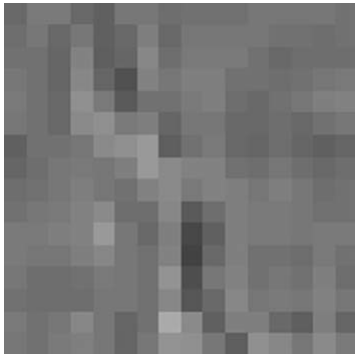
Hybrid DPCM/DCT coder

Example – encoder



Original luminance macroblock

Motion compensated prediction



Residual luminance macroblock

−4	−4	−1	0	1	1	0	−2
1	2	3	2	−1	−3	−6	−3
6	6	4	−4	−9	−5	−6	−5
10	8	−1	−4	−6	−1	2	4
7	9	−5	−9	−3	0	8	13
0	3	−9	−12	−8	−9	−4	1
−1	4	−9	−13	−8	−16	−18	−13
14	13	−1	−6	3	−5	−12	−7

Residual luminance samples

DCT

−13.50	20.47	20.20	2.14	−0.50	−10.48	−3.50	−0.62
10.93	−11.58	−10.29	−5.17	−2.96	10.44	4.96	−1.26
−8.75	9.22	−17.19	2.26	3.83	−2.45	1.77	1.89
−7.10	−17.54	1.24	−0.91	0.47	−0.37	−3.55	0.88
19.00	−7.20	4.08	5.31	0.50	0.18	−0.61	0.40
−13.06	3.12	−2.04	−0.17	−1.19	1.57	−0.08	−0.51
1.73	−0.69	1.77	0.78	−1.86	1.47	1.19	0.42
−1.99	−0.05	1.24	−0.48	−1.86	−1.17	−0.21	0.92

DCT coefficients of residual

Quantization

$$Q[DCT_i] = \text{round}\frac{DCT_i}{Q_{step}}$$

−1	2	2	0	0	−1	0	0
1	−1	−1	0	0	1	0	0
−1	1	−1	0	0	0	0	0
−1	−1	0	0	0	0	0	0
2	−1	0	0	0	0	0	0
−1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Quantized DCT coefficients

zig-zag

−1, 2, 1,−1,−1, 2, 0,−1, 1,−1, 2,−1,−1, 0, 0,−1, 0, 0, 0,−1,−1, 0, 0, 0, 0, 0, 1, 0,...

Array of reordered coefficients

RLE

(0,−1)(0, 2)(0, 1)(0,−1)(0,−1)(0, 2)(1,−1)(0, 1)(0,−1)(0, 2)
(0,−1)(0,−1)(2,−1)(3,−1)(0,−1)(5, 1)(EOB)

RLE code array

VLC

Run, Level, Last	VLC (+sign bit)
(0,−1, 0)	101
(0,2, 0)	11100
(0,1, 0)	100
(0,−1, 0)	101
(0,−1, 0)	101
(0,2, 0)	11100
(1,−1, 0)	1101
...	...
(5,1, 1)	00100110

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Hybrid DPCM/DCT coder

Example – decoder

−13.50	20.47	20.20	2.14	−0.50	−10.48	−3.50	−0.62
10.93	−11.58	−10.29	−5.17	−2.96	10.44	4.96	−1.26
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−1.99	−0.05	1.24	−0.48	−1.86	−1.17	−0.21	0.92

Original DCT coefficients

Error of lossy compression (quantization impact)

Entropy decode and reorder

−1	2	2	0	0	−1	0	0
1	−1	−1	0	0	1	0	0
−1	1	−1	0	0	0	0	0
−1	−1	0	0	0	0	0	0
2	−1	0	0	0	0	0	0
−1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Quantized DCT coefficients

Inverse quantization

$$R[DCT_i] = Q[DCT_i] \cdot Q_{step}$$

−12	24	24	0	0	−12	0	0
12	−12	−12	0	0	12	0	0
−12	12	−12	0	0	0	0	0
−12	−12	0	0	0	0	0	0
24	−12	0	0	0	0	0	0
−12	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Reconstructed DCT coefficients

IDCT

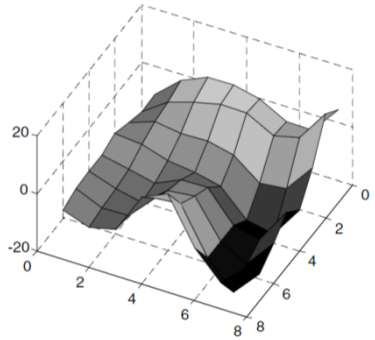
−4	−4	−1	0	1	1	0	−2
1	2	3	2	−1	−3	−6	−3
6	6	4	−4	−9	−5	−6	−5
10	8	−1	−4	−6	−1	2	4
7	9	−5	−9	−3	0	8	13
0	3	−9	−12	−8	−9	−4	1
−1	4	−9	−13	−8	−16	−18	−13
14	13	−1	−6	3	−5	−12	−7

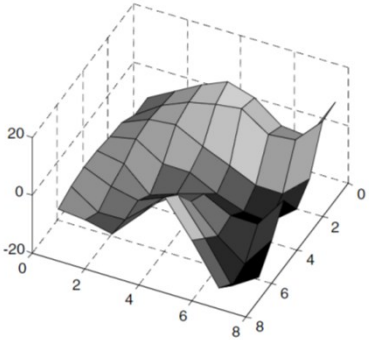
Reconstructed residual block

Error of lossy compression (quantization impact)

−3	−3	−1	1	−1	−1	−1	−3
5	3	2	0	−3	−4	−5	−6
9	6	1	−3	−5	−6	−5	−4
9	8	1	−4	−1	1	4	10
7	8	−1	−6	−1	2	5	14
2	3	−8	−15	−11	−11	−11	−2
2	5	−7	−17	−13	−16	−20	−11
12	16	3	−6	−1	−6	−11	−3

Original residual block





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Video preprocessing

Color sampling formats

- **Observer – lower sensitivity** to **details** in **color** than luma
  - **RGB representation** – all **components** with **identical resolution**
  - **YCbCr (YUV) representation** – **higher resolution** for **luma** than chroma

  - **Conversion between RGB and YCbCr**

$$Y = k_r R + (1 - k_b - k_r)G + k_b B$$

$$Cb = \frac{0.5}{1 - k_b}(B - Y), \qquad Cr = \frac{0.5}{1 - k_r}(R - Y)$$

$$G = Y - \frac{2k_b(1 - k_b)}{1 - k_b - k_r}Cb - \frac{2k_r(1 - k_r)}{1 - k_b - k_r}Cr$$

$$R = Y + \frac{1 - k_r}{0.5}Cr, \qquad B = Y + \frac{1 - k_b}{0.5}Cb$$

where e.g. for ITU-R BT.601 is $k_b = 0.114$ and $k_r = 0.299$

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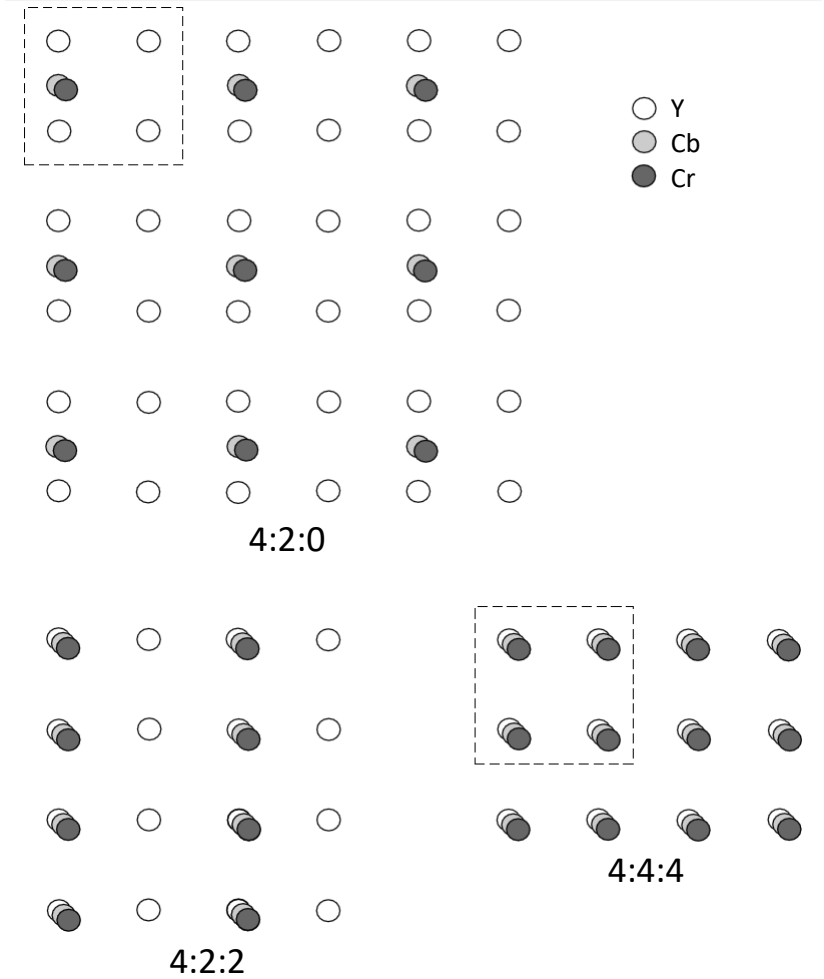
Video preprocessing

Color sampling formats

- Various **sampling formats** for **YCbCr**
  - **Most common** 4:4:4, 4:2:2, 4:2:0
  - **4:4:4 – full resolution** of all components
  - **4:2:2 – chroma** with **identical vertical resolution** as **chroma** but **half horizontal resolution**
  - **4:2:0 – chroma** has **half vertical** and **horizontal resolution**



Richardson, I. E. G.: H.264 and MPEG-4 Video Compression, Willey, 2003.

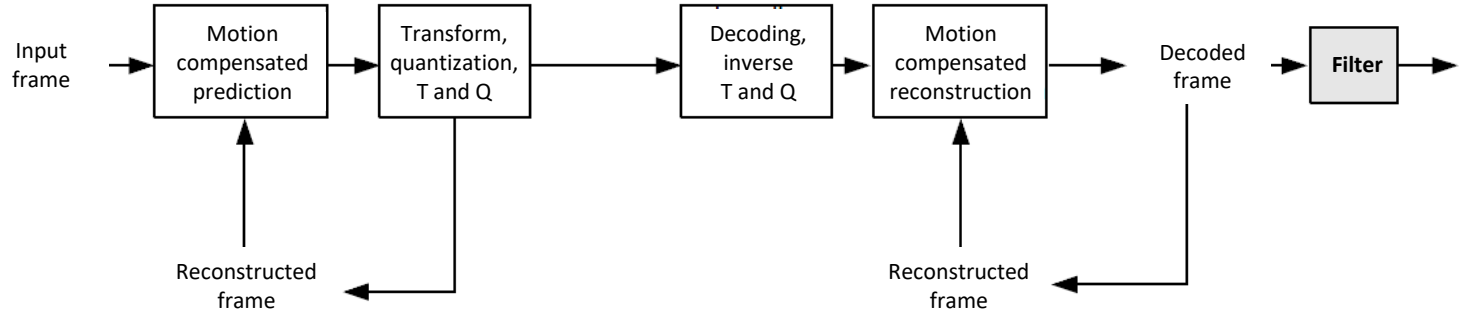
Video preprocessing

- **Improved efficiency** using **preprocessing techniques**
  - **Noise** in **video** (due to acquisition and digitization)
    - Reaching **higher spatial frequencies** and **variable in time**
    - Problem for **motion compensated prediction** and **quantization**
    - Causing **increase** of **required bitrate**
    - For **noisy video sequences** beneficial **noise suppressing filtration**
    - Parameters of **noise suppression** – **filtration** of **useful details prevented**
  - **Camera movements between frames** (camera shake)
    - Encoder treats movements as **global movement between frames**
    - Motion compensated prediction **not capable correcting large displacements**
    - Application of **image stabilization** (in camera or as postprocessing)
    - **Image stabilization** improves **compression efficiency**

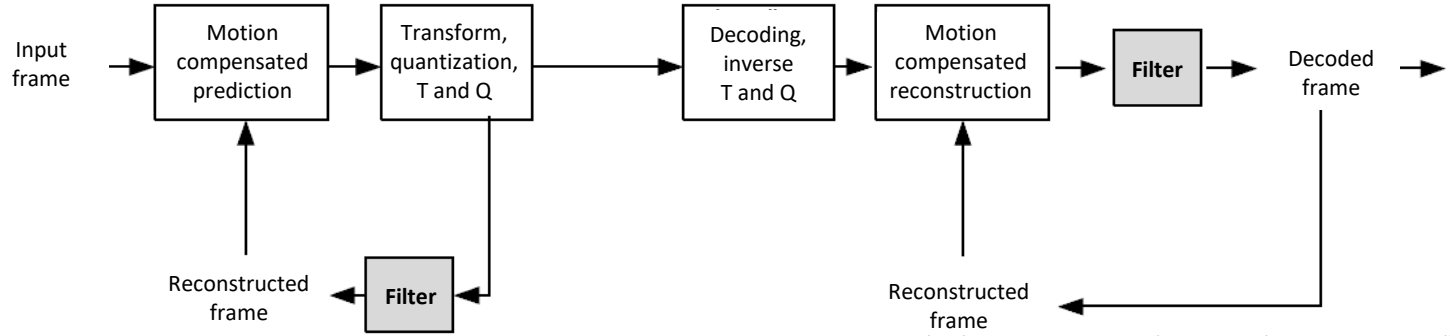
Video postprocessing

- **Improved efficiency** using **postprocessing filtration**
  - **Suppression** of **visible artifacts** (block structure)
    - **Filtered frames** can be used for **motion compensated prediction**
  - **De-blocking filter** (suppressing block structure)
    - **After decoder** (post-filter) or **inside decoder** (loop filter)
      - **Loop filter** part of **standard specification**

Postprocessing filtration (post-filter)



Filtration in coding loop (loop-filter)



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Recommended literature

- [1] Richardson, I. E. G.: **H.264 and MPEG-4 Video Compression**, Willey, 2003.
- [2] Bhaskaran, V., Konstantinides, K.: **Image and Video Compression Standards**, Kluwer, 1997.
- [3] Sayood, K.: **Introduction to Data Compression**, Morgan Kaufmann Publishers, 2006.
- [4] Gonzalez, R. C., Woods, R. E.: **Digital Image Processing**, Pearson, 2007.
- [5] Thyagarajan, K. S.: **Still Image and Video Compression with MATLAB**, Willey, 2011.

Thank you for your attention!

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